

A
Internship Report
on

“Data Analytics using Python”

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C E R T I F I C A T E

This is to certify that **Diksha Vinayak Satpute** has successfully completed the Internship entitled “**Data Analytics using Python**” under my supervision, in the partial fulfillment of Bachelor of Engineering (Computer) of Savitribai Phule Pune University, Pune.

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Thanking you,

Ms. Diksha Satpute

Principal

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To,
Diksha Satpute,
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We are pleased to inform you that you have been **selected for the position of Software Intern at Sumago Infotech Pvt. Ltd.**

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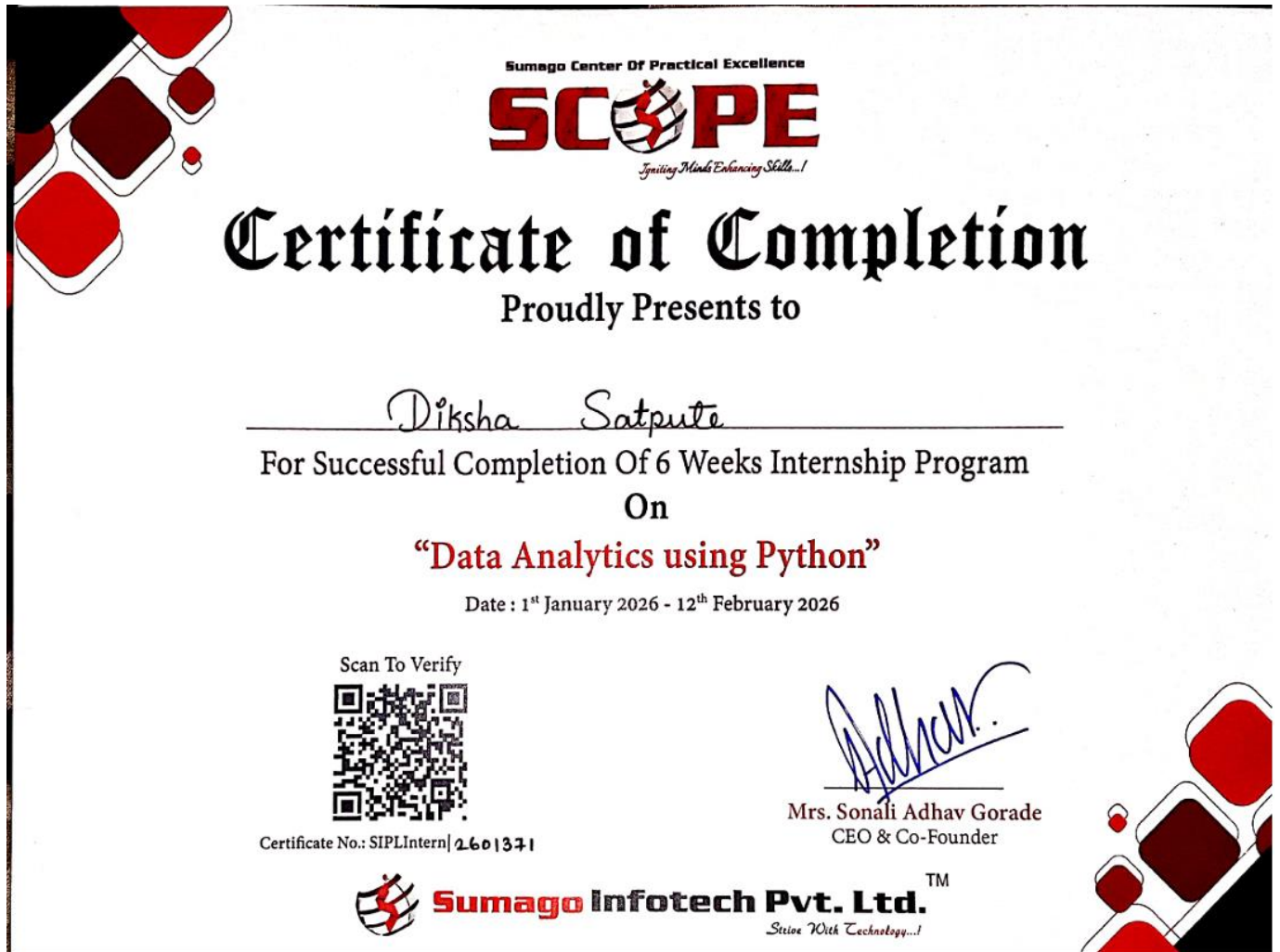
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ABSTRACT

Data analytics plays a vital role in converting raw and unstructured data into meaningful information that supports effective decision-making. In today's data-driven world, organizations rely heavily on data analytics to understand patterns, predict outcomes, and improve business performance. This project focuses on the application of data analytics using Python to analyse datasets and extract valuable insights that can help in understanding real-world problems. Python is chosen as the primary tool for this project because of its simplicity, flexibility, and strong community support. It offers a wide range of powerful libraries that make data analysis efficient and accurate. Libraries such as **Pandas** are used for data manipulation and handling structured data, **NumPy** is utilized for numerical computations, **Matplotlib** is used for basic data visualization, and **Seaborn** helps in creating advanced and visually appealing statistical charts.

The first stage of the project involves data collection and data pre-processing. Raw data often contains missing values, duplicate records, and inconsistent formats, which can affect the accuracy of the analysis. Therefore, data cleaning techniques are applied to handle missing values, remove errors, standardize data formats, and improve overall data quality. This step ensures that the dataset is reliable and suitable for further analysis. After data pre-processing, **Exploratory Data Analysis (EDA)** is performed to understand the structure, distribution, and behaviour of the data. EDA helps in identifying trends, relationships, outliers, and patterns within the dataset. Statistical methods such as mean, median, variance, and correlation are used along with graphical representations like bar charts, line graphs, histograms, scatter plots, and heatmaps. These visualizations make complex data easier to interpret and help in gaining meaningful insights at a glance. The project further demonstrates how Python can be effectively used for data analysis and visualization to support decision-making. By analysing the dataset, key factors influencing the data are identified, and important trends are highlighted. The insights obtained from the analysis can be used to make data-driven decisions, improve strategies, and predict future outcomes.

In conclusion, this project emphasizes the importance of data analytics in extracting useful information from large datasets. It also showcases Python as a reliable, efficient, and versatile tool for real-world data analysis tasks. Through proper data cleaning, analysis, and visualization, the project successfully demonstrates how meaningful insights can be generated to solve practical problems and support informed decision-making.

Keywords: Data Analytics, Python, EDA, Visualization, Data Cleaning, Pandas, NumPy

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LIST OF ABBREVIATIONS

Sr. No	Abbreviations	Illustration
1	Exploratory Data Analysis	EDA
2	Key Performance Indicator	KPI
3	Application Programming Interface	API
4	Structured Query Language	SQL
5	Data Analysis Expressions	DAX
6	Comma Separated values	CSV
7	JavaScript Object Notation	JSON
8	Business Intelligent	BI

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CHAPTER 1

INTRODUCTION

1.1 Background of Data Analytics

In the modern digital world, data has become one of the most valuable resources. With the rapid growth of technology, huge amounts of data are generated every day from various sources such as online shopping platforms, social media, banking systems, healthcare records, mobile applications, and sensors. This data includes customer details, transaction records, feedback, ratings, and behavioral patterns. However, raw data in its original form is often unstructured, incomplete, and difficult to understand.

Data analytics is the systematic process of analyzing this raw data to extract meaningful information. It involves collecting data, cleaning it, organizing it, and applying analytical techniques to identify patterns, relationships, and trends. The main purpose of data analytics is to convert raw data into useful insights that can support decision-making.

In earlier times, decisions were mainly based on experience and intuition. Today, organizations prefer **data-driven decisions**, where conclusions are drawn based on facts and evidence obtained from data analysis. This shift has made data analytics an essential part of almost every industry. As a result, data analytics has become a critical skill for students and professionals in the field of information technology, business, and management.

1.2 Importance of Data Analytics

The importance of data analytics lies in its ability to provide clarity and direction in complex situations. Organizations deal with massive datasets, and without proper analysis, this data cannot be used effectively. Data analytics helps in transforming this data into valuable insights that improve efficiency and performance.

One of the key benefits of data analytics is **improved decision-making**. Instead of relying on assumptions, organizations can analyze historical data and current trends to make accurate and informed decisions. Data analytics also helps in understanding customer behaviour, such as customer preferences, satisfaction levels, and buying patterns. This understanding enables companies to improve products and services.

Another important aspect of data analytics is **performance optimization**. By analysing operational data, organizations can identify inefficiencies, reduce costs, and improve productivity. Data analytics also plays a major role in risk management by identifying potential risks and anomalies early.

In academic projects, data analytics helps students develop logical thinking, problem-solving skills, and the ability to interpret data. It bridges the gap between theoretical knowledge and real-world applications.

1.3 Why Python for Data Analytics

Python is one of the most widely used programming languages in the field of data analytics. It is a high-level, open-source language known for its simplicity and readability. Python's syntax is easy to understand, making it an ideal choice for beginners as well as experienced professionals.

One of the major reasons for choosing Python for data analytics is its **rich ecosystem of libraries**. These libraries are specifically designed for handling data, performing calculations, and creating visualizations. Python allows analysts to perform complex tasks using fewer lines of code, which improves efficiency and reduces errors.

Python is also highly flexible and can work with different types of data formats such as CSV, Excel, JSON, and databases. Additionally, Python can be integrated with other tools and technologies, making it suitable for end-to-end data analysis.

Due to strong community support, Python users have access to extensive documentation, tutorials, and online resources. This makes learning and problem-solving much easier, especially for beginners.

1.4 Role of Python Libraries in Data Analytics

Python's effectiveness in data analytics largely depends on its powerful libraries. Each library plays a specific role in the data analytics process.

NumPy is used for numerical computations. It provides support for large multi-dimensional arrays and mathematical functions, making calculations faster and more efficient.

Pandas is one of the most important libraries for data analytics. It provides data structures like DataFrames and Series, which allow easy data manipulation, filtering, grouping, and aggregation. Pandas is mainly used for data cleaning and preprocessing.

Matplotlib is used for data visualization. It helps in creating charts such as line graphs, bar charts, histograms, and scatter plots. Visualization makes it easier to understand patterns and trends in data.

Seaborn is built on top of Matplotlib and provides more attractive and statistical

visualizations. It simplifies the process of creating complex graphs and improves visual appeal. Together, these libraries make Python a complete and powerful tool for Analytics

1.5 Data Analytics Process Using Python

The data analytics process follows a structured approach to ensure accuracy and meaningful results.

1.5.1 Data Collection

Data collection is the first step in the analytics process. Data can be collected from various sources such as files, databases, or online platforms. The quality and relevance of the data collected directly affect the outcome of the analysis.

1.5.2 Data Cleaning and Preprocessing

Raw data often contains missing values, duplicate records, and incorrect entries. Data cleaning involves handling these issues to improve data quality. Preprocessing ensures that the data is consistent and suitable for analysis.

1.5.3 Exploratory Data Analysis (EDA)

EDA is performed to understand the data better. Statistical measures and visualizations are used to identify trends, patterns, and relationships. This step helps in forming assumptions and guiding further analysis.

1.5.4 Data Visualization

Visualization helps in presenting data insights clearly and effectively. Graphs and charts make complex data easy to understand for both technical and non-technical users.

1.5.5 Insight Generation

Based on analysis and visualization, meaningful insights are derived. These insights help in understanding the problem better and making recommendations.

1.5.6. Applications of Data Analytics Using Python

Data analytics using Python is widely used across different domains. In business, it helps in sales analysis and customer segmentation. In healthcare, it is used for patient data analysis and disease prediction. In finance, it supports fraud detection and risk assessment. In e-commerce, it is used for recommendation systems and review analysis. These applications show the wide usability of Python in data analytics.

1.6 Objective of the Project

The main objective of this project is to understand and implement data analytics techniques using Python in order to extract meaningful insights from raw data. The project aims to provide practical exposure to the complete data analytics process, starting from data collection and cleaning to analysis, visualization, and interpretation of results. By working on real or realistic datasets, the project helps in understanding how data analytics supports effective decision-making in real-world scenarios.

One of the key objectives is to analyze datasets using Python libraries such as Pandas and NumPy. Through data analysis, the project focuses on identifying patterns, trends, correlations, and anomalies present in the data. This helps in gaining a deeper understanding of the dataset and the factors influencing it. The analytical process enables the transformation of raw data into structured and meaningful information.

Another important objective of the project is to visualize insights using graphs and charts. Data visualization plays a crucial role in simplifying complex data and presenting findings in an easy-to-understand manner. Using visualization libraries such as Matplotlib and Seaborn, the project aims to represent data through bar charts, line graphs, histograms, scatter plots, and heatmaps. These visualizations make it easier to communicate insights to both technical and non-technical audiences.

The project also aims to derive conclusions and insights that support data-driven decision-making. By interpreting analytical results and visual outputs, the project highlights key findings that can help in improving strategies, identifying opportunities, and addressing potential issues. This objective emphasizes the importance of using data as a reliable basis for making informed decisions rather than relying on assumptions.

In addition, the project focuses on building a strong foundation in data analytics concepts and Python programming. It helps in developing essential skills such as data cleaning, exploratory data analysis, logical thinking, and problem-solving. The project also improves proficiency in Python programming by applying coding concepts to practical data analytics tasks.

Overall, the project aims to bridge the gap between theoretical knowledge and practical implementation of data analytics. It equips learners with the necessary technical skills and analytical mindset required to perform real-world data analysis and prepares them for future roles in the field of data analytics.

1.7. Conclusion of Introduction

In conclusion, data analytics plays a crucial role in converting raw and unorganized data into meaningful and useful insights that support informed decision-making. In today's digital era, the ability to analyze data effectively has become essential across various industries. Data analytics helps organizations and individuals understand patterns, trends, and relationships within data, enabling better planning, improved performance, and strategic growth.

Python emerges as an effective and reliable tool for data analytics due to its simplicity, flexibility, and extensive range of powerful libraries. Libraries such as Pandas, NumPy, Matplotlib, and Seaborn allow efficient data handling, analysis, and visualization. Python's user-friendly syntax and strong community support make it suitable for both beginners and advanced learners, enabling them to perform complex data analytics tasks with ease.

This project provides valuable practical exposure to fundamental data analytics concepts, including data cleaning, exploratory data analysis, visualization, and interpretation of results. By working with real-world datasets and applying Python-based techniques, learners gain hands-on experience that bridges the gap between theoretical knowledge and practical implementation.

Overall, the project lays a strong foundation for understanding data analytics using Python and prepares learners for advanced studies and career opportunities in data science and analytics. It helps in developing analytical thinking, problem-solving skills, and technical proficiency, which are essential for successfully working in data-driven environments.

Chapter 2

INTERNSHIP PROJECT

2.1 Introduction to project :-

In today's digital era, online shopping platforms generate a huge amount of data every day. This data includes information about products, customer preferences, ratings, pricing, sales trends, and customer reviews. Analysing this data helps companies understand consumer behaviour, improve their products, and make better business decisions. The objective of the **Online Product Analysis project** was to analyse product data collected from an online marketplace and extract useful insights from it. By analysing product performance, ratings, pricing patterns, and category-wise trends, businesses can identify which products perform well in the market and which ones need improvement. To perform this analysis effectively, we used **Power BI**, a powerful data visualization and business intelligence tool developed by **Microsoft**. Power BI allows users to transform raw data into interactive dashboards and visual reports that help in understanding complex data easily. During the internship, this project provided practical exposure to data analytics concepts such as **data cleaning, data transformation, data visualization, and business insight generation**.

In today's digital era, online shopping platforms have become one of the most important components of the global economy. E-commerce websites such as Amazon, Flipkart, and other online marketplaces generate an enormous amount of data every day through customer interactions, product listings, transactions, and reviews. Every time a customer searches for a product, compares prices, reads reviews, or makes a purchase, valuable data is generated and stored in the system. This data includes various types of information such as product names, product categories, prices, discounts, customer ratings, number of reviews, sales volume, and customer purchasing patterns. The availability of such large volumes of data creates opportunities for businesses to analyse and understand customer behaviour more effectively. By analysing product-related data, companies can identify which products are most popular among customers, which categories generate higher demand, and how pricing strategies influence purchasing decisions. For example, products with higher ratings and positive customer reviews are generally more trusted and are likely to attract more buyers. Similarly, analysing price ranges and discount patterns helps companies understand how customers respond to promotional offers and seasonal sales. Therefore, analysing online product data is extremely important for businesses as it enables them to improve product quality, optimize pricing strategies, enhance customer satisfaction, and increase overall

sales performance. The main objective of the **Online Product Analysis project** undertaken during the internship was to analyse a dataset containing information about products listed on an online marketplace and extract meaningful insights from it. The project aimed to study different aspects of product performance such as ratings, pricing patterns, product categories, and customer feedback. By examining these factors, the analysis helps identify trends and patterns that influence customer purchasing behaviour. For instance, the analysis can highlight which product categories have the highest number of listings, which products receive the best ratings from customers, and which price ranges are most common in the market. Such insights are valuable for businesses as they support better decision-making in areas such as product development, marketing strategies, and inventory management. To perform this analysis effectively, the project utilized **Power BI**, which is a powerful business intelligence and data visualization tool developed by Microsoft. Power BI is widely used in the field of data analytics because it allows users to easily import, transform, analyse, and visualize large datasets. One of the key advantages of Power BI is its ability to convert raw and unstructured data into meaningful visual representations such as charts, graphs, tables, and dashboards. These visualizations make it easier to interpret complex data and identify patterns or trends that might not be immediately visible in raw numerical data. Power BI also provides powerful features such as **data cleaning, data transformation, and interactive reporting**. Before performing any analysis, the raw dataset often contains inconsistencies such as missing values, duplicate records, or incorrect data formats. Using Power Query, which is a built-in data preparation tool in Power BI, these issues can be corrected by removing duplicates, handling missing data, renaming columns, and converting data types. This step ensures that the dataset becomes accurate, consistent, and suitable for further analysis. Another important feature of Power BI is its ability to create **interactive dashboards**. These dashboards allow users to explore the data dynamically by applying filters, slicers, and drill-down features. For example, a user can select a particular product category and instantly view information such as the average rating, price distribution, and number of reviews related to that category. This level of interactivity makes Power BI an extremely useful tool for business analysis and decision-making. During the internship, working on the Online Product Analysis project provided valuable hands-on experience in several important areas of data analytics. It helped in understanding the complete workflow of a data analysis project, starting from **data collection and cleaning to visualization and**

interpretation of insights. The project also enhanced practical skills in using Power BI tools for building dashboards and presenting analytical findings in a clear and visually appealing manner.

Overall, the project demonstrated how data analytics can be used to transform raw data into meaningful insights that help businesses understand market trends, improve customer experience, and make informed business decisions. It also provided practical exposure to modern business intelligence tools and techniques that are widely used in industries today.

2.2 About the software

Power BI is a powerful business analytics and data visualization tool developed by Microsoft that helps organizations analyse large amounts of data and convert it into meaningful insights. In modern businesses, large volumes of data are generated every day from different sources such as online transactions, customer interactions, and digital platforms. Analysing this data manually is difficult and time-consuming. Power BI simplifies this process by allowing users to collect, process, analyse, and visualize data in an interactive and user-friendly manner.

The main purpose of Power BI is to transform raw and unstructured data into useful business insights that help organizations make informed decisions. It enables users to create interactive reports, dashboards, and visualizations that clearly represent patterns, trends, and relationships within the data. Instead of analysing large spreadsheets with thousands of rows of numbers, Power BI presents the information in visual formats such as graphs and charts, making it easier to interpret and understand.

One of the major advantages of Power BI is its ability to connect to multiple data sources. It supports data integration from various platforms including Excel files, SQL databases, cloud-based storage systems, web services, and APIs. This flexibility allows analysts to combine data from different sources into a single platform for comprehensive analysis. After importing the data, Power BI provides tools to clean and transform the dataset so that it becomes structured and suitable for analysis. Another important feature of Power BI is its interactive nature. Users can apply filters, slicers, and drill-down features to explore the data in more detail. For example, if a dashboard shows product sales across different categories, a user can select a particular category and instantly view more detailed information related

to that category. This makes Power BI very useful for business intelligence and decision-making.

2.3 Main Components of Power BI

Power BI consists of several components that work together to perform data analysis and visualization.

1. Power BI Desktop

Power BI Desktop is the primary application used for building reports and dashboards. It is a Windows-based software that allows users to connect to different data sources, transform the data, and create visual reports.

In Power BI Desktop, users can perform multiple operations such as importing datasets, editing and cleaning data, designing data models, and creating visualizations. It provides a user-friendly interface with drag-and-drop functionality, which makes it easier for beginners to create analytical reports without extensive programming knowledge.

Power BI Desktop contains three main views:

- Report View – Used to create visualizations and dashboards.
- Data View – Used to examine the dataset in a tabular format.
- Model View – Used to manage relationships between tables.

These views help users understand and organize the data effectively during the analysis process.

2. Power Query Editor

Power Query Editor is an important component of Power BI that is used for data cleaning and data transformation. Raw datasets often contain errors such as missing values, duplicate records, inconsistent formatting, or unnecessary columns. These issues must be corrected before performing analysis.

Using Power Query Editor, several data preparation tasks can be performed, such as:

- Removing duplicate rows
- Handling missing values
- Splitting or merging columns
- Renaming columns
- Changing data types (for example text to numeric values)
- Filtering unnecessary records

Power Query follows a step-by-step transformation process, meaning every change

applied to the dataset is recorded as a step. This makes the data preparation process transparent and repeatable. Once the data is cleaned and transformed, it is loaded into Power BI for further analysis.

3. Data Model

The data modeling feature in Power BI helps organize and structure the dataset in a way that improves performance and accuracy during analysis. When multiple datasets are used, relationships can be created between tables based on common fields such as product ID, category ID, or date.

For example, one table may contain product details while another table contains product reviews or sales data. By creating relationships between these tables, Power BI can combine the data and perform more meaningful analysis.

Data modeling also allows the creation of calculated columns and measures using the Data Analysis Expressions (DAX) language. These calculations help generate additional insights such as total sales, average ratings, or percentage growth.

4. Data Visualization

One of the most powerful features of Power BI is its ability to create interactive data visualizations. Visualizations help convert complex datasets into graphical representations that are easier to understand and analyse.

Power BI provides a wide range of visualization tools such as:

- Bar Charts – Used to compare values across different categories.
- Line Charts – Used to display trends over time.
- Pie Charts – Used to represent percentage distribution of data.
- Tables and Matrices – Used to display detailed numerical information.
- KPI Indicators – Used to track important business performance metrics.
- Cards – Used to highlight key numerical values such as total sales or average rating.
- Filters and Slicers – Used to allow users to interactively filter the data.

These visual tools make it easier to identify patterns, trends, and relationships within the data. For example, a bar chart can quickly show which product category has the highest number of listings, while a line chart can reveal changes in sales trends over time.

5. Dashboards and Reports

The final output of Power BI analysis is usually presented in the form of reports and

dashboards. A report consists of multiple pages of visualizations that provide detailed insights into the dataset. A dashboard, on the other hand, is a consolidated view that displays the most important metrics and visualizations in a single screen.

Dashboards are designed to be interactive and user-friendly. Users can apply filters, select specific categories, and drill down into detailed information. This interactivity allows decision-makers to explore the data dynamically and gain deeper insights.

For example, in an online product analysis dashboard, users can view information such as top-rated products, product category distribution, average product ratings, price comparisons, and discount patterns. These insights help businesses understand market trends and improve their strategies accordingly.

Overall, Power BI plays a crucial role in modern data analytics by enabling users to transform raw data into interactive and visually appealing reports. Its powerful data integration, transformation, and visualization capabilities make it one of the most widely used business intelligence tools in industries today.

2.4 Objectives of Project

The Online Product Analysis Project was undertaken with the aim of analysing product-related data obtained from an online marketplace and extracting meaningful insights from it. In today's competitive digital marketplace, businesses rely heavily on data to understand market trends, customer preferences, and product performance. The objective of this project was to utilize data analytics techniques and visualization tools to study different aspects of online product data and provide insights that could help businesses make informed decisions. One of the primary objectives of the project was to analyse online product sales data. Online marketplaces generate large volumes of data related to product listings, sales performance, customer interactions, and purchasing behaviour. By analysing this data, it becomes possible to identify patterns such as which products are selling more frequently, which categories are gaining popularity, and which products may require improvement or better marketing strategies. Understanding sales data helps businesses track product demand and make decisions related to inventory management and marketing. Another important objective of the project was to understand customer preferences based on ratings and reviews. Customer ratings and feedback are essential indicators of product quality and customer satisfaction. Products that receive higher ratings and positive reviews generally

attract more buyers, while products with lower ratings may indicate issues related to quality, usability, or customer expectations. By analysing ratings and review counts, businesses can identify products that are performing well and those that need improvement. This analysis also helps companies understand what features customers value the most. The project also aimed to identify the best-selling or most popular products available on the platform. By analysing data such as the number of reviews, ratings, and product distribution, it becomes possible to determine which products are performing well in the market. Identifying these products helps businesses understand successful product strategies and focus more on items that generate higher demand and customer interest. Another key objective was to analyse product categories and their overall performance. Online marketplaces typically contain multiple product categories such as electronics, clothing, home appliances, and accessories. By analysing the distribution of products across categories, it is possible to understand which categories have the highest number of products and which categories receive better customer ratings. This information helps businesses identify strong market segments and areas with potential for growth. The project also focused on studying pricing patterns and the effects of discounts on products. Pricing plays a significant role in influencing customer purchasing decisions. Customers often compare prices across different products before making a purchase. Additionally, discounts and promotional offers can significantly impact product popularity and sales performance. By analysing product prices and discount percentages, the project aimed to understand how pricing strategies affect customer behaviour and product demand. Another important objective of the project was to create an interactive dashboard using Power BI for effective data visualization. Data visualization plays a crucial role in data analytics because it helps convert complex numerical data into easily understandable graphical representations. Instead of analysing large spreadsheets of raw data, dashboards present insights through charts, graphs, and visual indicators. The Power BI dashboard created during this project allowed users to explore product data through visualizations such as bar charts, pie charts, and tables. It also included interactive features like filters and slicers, enabling users to focus on specific product categories or metrics. Overall, the main goal of this project was to transform raw and unstructured product data into meaningful insights that could support data-driven decision-making. By analysing product performance, customer preferences, and pricing trends, the project demonstrated how businesses can use data analytics tools to better understand market

behaviour and improve their strategic planning. The insights generated through this analysis can help companies optimize their product offerings, enhance customer satisfaction, and improve overall business performance.

2.5Dataset description

The dataset used in this project contained information related to products listed on an online shopping platform. The dataset included several important attributes such as:

- Product Name
- Product Category
- Product Price
- Discount Percentage
- Product Rating
- Number of Reviews
- Brand Name
- Product Availability

This dataset helped in analysing how different factors such as price, ratings, and reviews influence product popularity and sales performance.

The dataset used in the Online Product Analysis Project contained detailed information about various products listed on an online shopping platform. In data analytics projects, the dataset plays a very important role because it provides the raw information that is analysed to generate meaningful insights. The dataset used in this project consisted of multiple records representing different products available on the platform, along with several attributes describing their characteristics and performance. Each row in the dataset represented a single product, while each column represented a specific attribute related to that product. These attributes provided valuable information such as the product's name, category, price, discount percentage, customer ratings, number of reviews, brand information, and product availability. By analysing these attributes collectively, it becomes possible to understand how different factors influence product popularity and customer purchasing behaviour. One of the key attributes in the dataset was the Product Name. This column contained the name or title of the product listed on the online platform. The product name helps identify the item being analysed and is useful when comparing the performance

of different products. It also helps categorize products based on type or functionality. For example, the dataset may contain product names related to electronics, clothing, home appliances, or accessories.

Another important attribute was the Product Category. Product categories group similar products together based on their type or function. Examples of product categories include electronics, fashion, home and kitchen appliances, beauty products, and sports equipment. Analysing product categories helps understand which categories contain the highest number of products and which categories perform better in terms of ratings and customer engagement. This information is useful for businesses when planning inventory, marketing campaigns, and product development strategies. The dataset also included Product Price, which represents the actual cost of the product listed on the platform. Price is a critical factor that influences customer purchasing decisions. Customers often compare prices across similar products before making a purchase. By analysing product prices across different categories, businesses can identify pricing patterns, determine average price ranges, and understand how pricing impacts product popularity and ratings. Another important attribute in the dataset was Discount Percentage. Online shopping platforms frequently offer discounts to attract customers and increase sales. The discount percentage indicates how much price reduction is offered on a particular product compared to its original price. Analysing discount data helps determine whether products with higher discounts attract more customer attention or generate more reviews. It also helps businesses understand the effectiveness of promotional strategies and seasonal sales campaigns. The dataset also contained Product Ratings, which represent the average rating given by customers after purchasing or using the product. Ratings are usually provided on a scale, such as 1 to 5 stars. Higher ratings generally indicate better customer satisfaction and product quality. Analysing product ratings helps identify which products are highly appreciated by customers and which products may require improvements. It also provides insight into customer trust and product reliability. Another important attribute was the Number of Reviews. Customer reviews provide detailed feedback about a product's performance, quality, and usability. The number of reviews indicates how many customers have shared their opinions about the product. Products with a higher number of reviews are usually more popular or widely purchased. Analysing review counts helps identify trending products and understand the level of customer engagement with different products. The Brand Name attribute represents the company or brand that manufac-

tures or sells the product. Brand reputation plays an important role in influencing customer purchasing decisions. Well-known brands often receive higher trust from customers and may have better ratings or more reviews. By analysing brand information, businesses can identify which brands perform better in the market and which brands receive higher customer satisfaction. Another attribute included in the dataset was Product Availability. This indicates whether a product is currently available for purchase on the platform or is out of stock. Availability information helps businesses understand inventory status and demand for certain products. If a product frequently goes out of stock, it may indicate high demand among customers. Overall, the dataset provided a comprehensive view of different aspects of online products, including pricing, customer feedback, and product characteristics. By analysing these attributes collectively using Power BI, it becomes possible to identify trends, patterns, and relationships between different factors. For example, the analysis can reveal whether products with higher discounts receive more reviews, whether certain categories receive better ratings, or whether well-known brands attract more customers.

Therefore, the dataset served as the foundation for the entire analysis process and helped transform raw product data into meaningful insights that can support better business decisions in the online marketplace.

2.6 Steps to make project dashboard

The Online Product Analysis project was completed in several steps.

Step 1: Data Collection The first step involved collecting the dataset containing product information. The dataset was provided in Excel format, which included various product attributes such as pricing, ratings, and categories.

Step 2: Importing Data into Power BI

After obtaining the dataset, the next step was to import it into Power BI Desktop.

Steps followed:

1. Open Power BI Desktop.
2. Click on Get Data.
3. Select Excel file.
4. Upload the dataset.

5. Load the data into Power BI.

Once the data was loaded, it appeared in the data view where we could see all the columns and rows.

Step 3: Data Cleaning and Transformation

Before analysing the data, it was necessary to clean and organize it using Power Query Editor.

Data cleaning steps included:

- Removing duplicate records.
- Handling missing values.
- Renaming columns for better understanding.
- Converting data types (for example, text to numeric).
- Removing unnecessary columns.

This step ensured that the dataset was accurate and ready for analysis.

Step 4: Data Modeling

In this step, relationships between different data tables were created (if multiple tables were used). This helps Power BI understand how data is connected.

For example:

- Product categories linked with product details.
- Ratings linked with product IDs.

Data modeling helps improve the performance and accuracy of reports.

Step 5: Creating Visualizations

Once the data was cleaned and structured, various visualizations were created to analyse product performance.

Some of the key visualizations created were:

1. Category-wise Product Distribution

A bar chart was used to show the number of products available in each category. This helped identify which product categories had the highest number of products.

2. Product Rating Analysis

A column chart was used to analyse average ratings of products across different categories.

This helped understand customer satisfaction levels.

3. Price Distribution

A histogram or bar chart was used to analyse price ranges of products and identify affordable vs premium products.

Next points indeed

4. Discount Analysis

A pie chart was used to show the distribution of discount percentages offered on different products.

5. Top Rated Products

A table visualization was created to show the top-rated products based on customer ratings and reviews.

Step 6: Creating Dashboard

After creating individual visualizations, they were combined into a single interactive dashboard.

The dashboard included:

- Product category distribution
- Top selling products
- Average ratings
- Price comparison
- Discount trends

Power BI allows users to apply filters and slicers, which means the user can select a category or brand and instantly see the updated analysis.

This makes the dashboard interactive and user-friendly.

2.7 Key insight :-

After performing the analysis of the dataset using Power BI, several meaningful insights were obtained regarding product performance, customer behaviour, and market trends in the online shopping platform. These insights were derived from different visualizations such as bar charts, pie charts, tables, and category-wise comparisons created in the Power BI dashboard. Analysing these insights helps businesses understand how different factors such as product category, price, ratings, reviews, and discounts influence customer decisions and overall product performance. One of the key insights

observed from the analysis was that **certain product categories had significantly more listings than others**. When the dataset was analysed based on product categories, it was found that some categories contained a large number of products compared to others. This indicates that these categories are highly competitive and have a larger variety of products available for customers to choose from. Categories with more product listings usually represent high-demand markets where many sellers are competing to attract customers. On the other hand, categories with fewer listings may indicate niche markets or product areas with lower competition. Understanding the distribution of products across categories helps businesses identify market opportunities and determine which product segments have the greatest potential for growth. Another important insight obtained from the analysis was that **products with higher ratings and a greater number of reviews were generally more popular among customers**. Customer ratings and reviews play a crucial role in influencing purchasing decisions. When customers shop online, they often rely on the experiences of previous buyers to evaluate the quality and reliability of a product. Products with high ratings are perceived as trustworthy and are more likely to attract new customers. Similarly, a higher number of reviews indicates that many customers have interacted with the product, which increases its credibility. The analysis showed that products with both high ratings and a large number of reviews tended to receive more attention and engagement from customers. The analysis also revealed that **discounted products attracted more customer attention compared to products with little or no discount**. Online shopping platforms frequently use discount strategies to attract buyers and encourage purchases. Products offering higher discounts often appear more appealing to customers because they provide perceived value for money. The data analysis showed that products with noticeable discounts often received higher review counts and engagement levels. This indicates that promotional offers and discounts play an important role in influencing customer behaviour and can significantly impact product popularity. Another key observation from the analysis was that **some brands consistently received better ratings than others**. Brand reputation is an important factor that affects customer trust and product perception. Well-established brands often maintain better quality standards and customer satisfaction levels, which leads to higher ratings and positive feedback from customers. The dataset analysis revealed that certain brands had consistently high ratings across multiple products. This suggests that customers tend to trust and prefer products from brands with a strong reputation in the market. Businesses can use this insight to focus on maintaining brand quality and strengthening customer trust. The analysis also highlighted that **price range had a significant influence on customer ratings and purchasing behaviour**. Pricing plays an important role in shaping customer decisions. Some customers prefer affordable products that offer good value for money, while others may prefer premium products with higher quality and features. By analysing product prices and ratings, it was observed that certain price ranges attracted more customer en-

gement and reviews. Products priced competitively often received more attention because they balanced affordability with quality. This insight helps businesses understand the optimal pricing strategies required to attract customers while maintaining profitability. Overall, these insights demonstrate how different factors such as product categories, ratings, reviews, discounts, brand reputation, and pricing interact to influence product performance in an online marketplace. By analysing these elements together, businesses can gain a deeper understanding of customer preferences and market behaviour.

These insights can help companies improve their **product development strategies, marketing campaigns, and pricing policies**. For example, businesses can focus on offering competitive discounts, improving product quality to achieve higher ratings, and strengthening brand reputation to build customer trust. Additionally, understanding category-wise performance can help companies identify high-demand product segments and allocate resources accordingly.

Therefore, the analysis not only provided valuable insights into product performance but also demonstrated how data analytics tools like Power BI can support **data-driven decision-making** in modern business environments.

2.8 Advantages of power Bi

Using **Power BI** for the Online Product Analysis project provided several important advantages that made the process of analysing and interpreting data easier and more efficient. Power BI is widely used in the field of data analytics and business intelligence because it offers powerful tools for data processing, visualization, and reporting. These features allow analysts to convert raw data into meaningful insights that support better business decisions. One of the major advantages of Power BI is **easy data visualization**. In data analysis, large datasets often contain thousands of rows and columns of numerical information, which can be difficult to understand when viewed in a spreadsheet format. Power BI helps solve this problem by converting raw numerical data into graphical representations such as charts, graphs, and tables. Visualizations such as bar charts, pie charts, and line graphs allow users to quickly identify patterns, trends, and comparisons within the data. For example, in the Online Product Analysis project, product categories, ratings, and pricing patterns could be easily understood through visual charts instead of manually analysing numbers. Another important advantage of Power BI is the ability to

create **interactive dashboards**. Dashboards allow multiple visualizations to be displayed together on a single screen, providing a comprehensive overview of the data. These dashboards are interactive, meaning users can apply filters, select specific categories, and explore the data in more detail. For instance, a user can select a particular product category and instantly view related information such as product ratings, price distribution, and number of reviews. This interactive functionality makes the analysis more dynamic and user-friendly compared to static reports.

Power BI also has the **ability to handle large datasets efficiently**. Modern businesses generate huge amounts of data every day, and analysing this data requires tools that can process large volumes of information quickly. Power BI is designed to manage and analyse large datasets without significantly affecting performance. It allows users to perform calculations, create relationships between data tables, and generate insights even when working with complex datasets. Another key advantage is **real-time data analysis**. Power BI supports real-time data updates when connected to live data sources. This means that whenever the underlying data changes, the visualizations and reports can automatically update to reflect the latest information. This feature is particularly useful for businesses that need to monitor performance metrics continuously, such as sales trends, customer engagement, or product demand. Power BI also provides strong **integration with multiple data sources**. It can connect to a wide variety of platforms including Excel files, SQL databases, cloud storage services, and web-based APIs. This allows analysts to collect data from different sources and combine them into a single platform for comprehensive analysis. In the Online Product Analysis project, the dataset could be easily imported and processed within Power BI, allowing all analysis and visualization tasks to be performed within one environment.

Overall, Power BI simplifies complex data analysis by providing powerful tools for data preparation, visualization, and reporting. It enables users to identify patterns, trends, and relationships within the data more efficiently. As a result, businesses can better understand market behaviour, customer preferences, and product performance, ultimately helping them make more informed and data-driven decisions.

Chapter 3

ACTUAL IMAGES

3.1 First Image of Project Dashboard



Figure 3.1: Online Product Review Analysis – Dashboard Overview

This page represents the main dashboard interface of the Online Product Review Analysis project. It acts as the introductory screen that provides a visual overview of the entire analytical system developed using Power BI. The dashboard integrates multiple product categories such as electronics, clothing, home appliances, sports equipment, and other consumer goods, representing the wide range of products analysed in the dataset.

The purpose of this page is to provide users with a high-level understanding of the project theme and the type of products included in the analysis. The visual elements displayed on the dashboard represent various product categories commonly found on online shopping platforms. These elements help illustrate the scope of the analysis, which includes multiple product types from different categories. Additionally, the dashboard includes location indicators representing different cities such as Aurangabad, Nashik, Bengaluru, Pune, Delhi, Mumbai, Hyderabad, and Nagpur. These locations are used in the analysis to compare customer reviews and ratings across different geographical areas. By incorporating city-wise data, the project aims to analyse how customer feedback may vary depending on location. Overall, this introductory dashboard provides a visual summary of the project and sets the foundation for the subsequent analytical pages, which focus on product reviews, customer feedback, and delivery performance.

3.2 Second page of the Dashboard

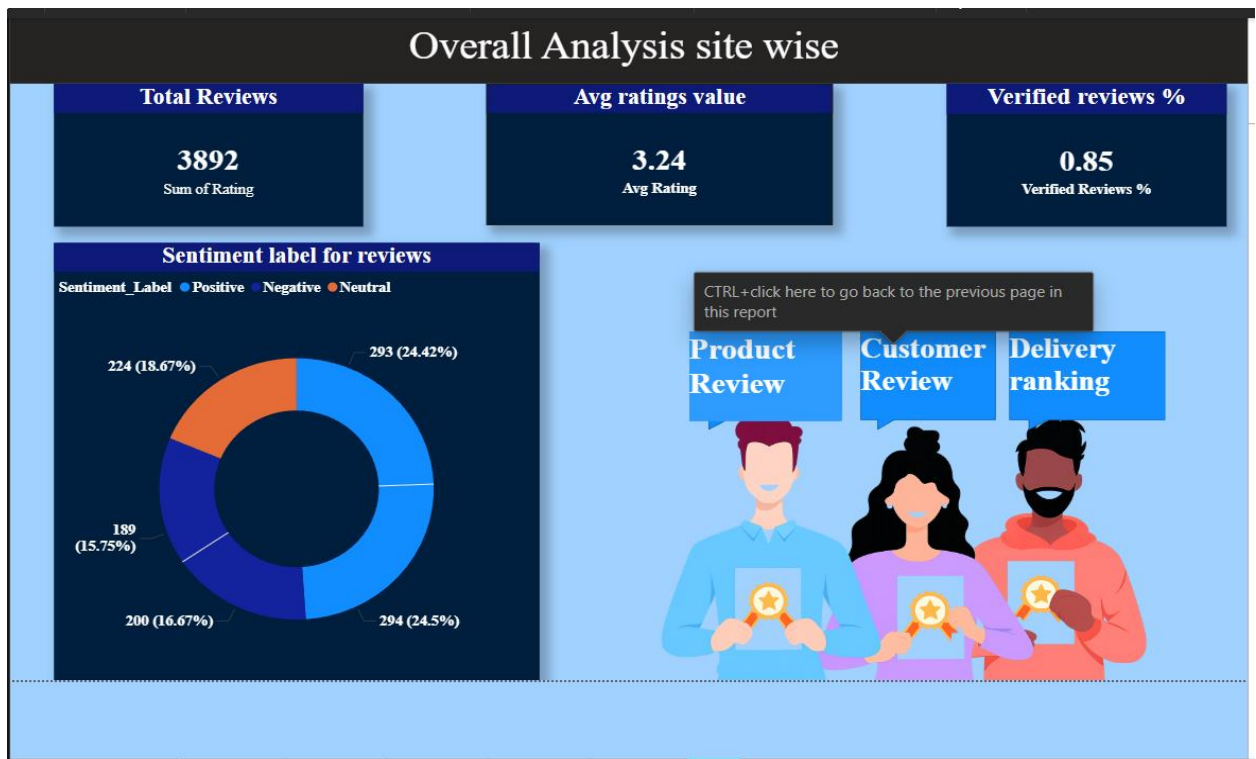


Figure 3.2: Overall Analysis – Site Wise Review Insights

This dashboard provides an overall summary of customer reviews and ratings collected from the dataset. It presents key performance indicators (KPIs) that give a quick overview of customer feedback and sentiment. The dashboard displays three main summary metrics:

Total Reviews: The total number of reviews recorded in the dataset is 3892, representing the overall customer feedback collected for different products across the platform. This metric helps understand the overall level of customer engagement with the products.

Average Rating Value: The average rating across all products is 3.24, indicating that the overall customer satisfaction level is moderate. This suggests that while some products receive positive feedback, there are also products that require improvements in quality, service, or delivery.

Verified Reviews Percentage: The verified reviews percentage is 0.85 (85%), which indicates that a large portion of the reviews are from verified purchases. Verified reviews are generally considered more reliable because they come from customers who have actually purchased the product.

The dashboard also includes a Sentiment Analysis visualization, which categorizes reviews into three types: Positive, Negative, Neutral.

This analysis helps understand customer emotions and opinions about products. The chart shows the distribution of sentiments across all reviews, allowing businesses to identify the.

3.3 Third page of the project dashboard

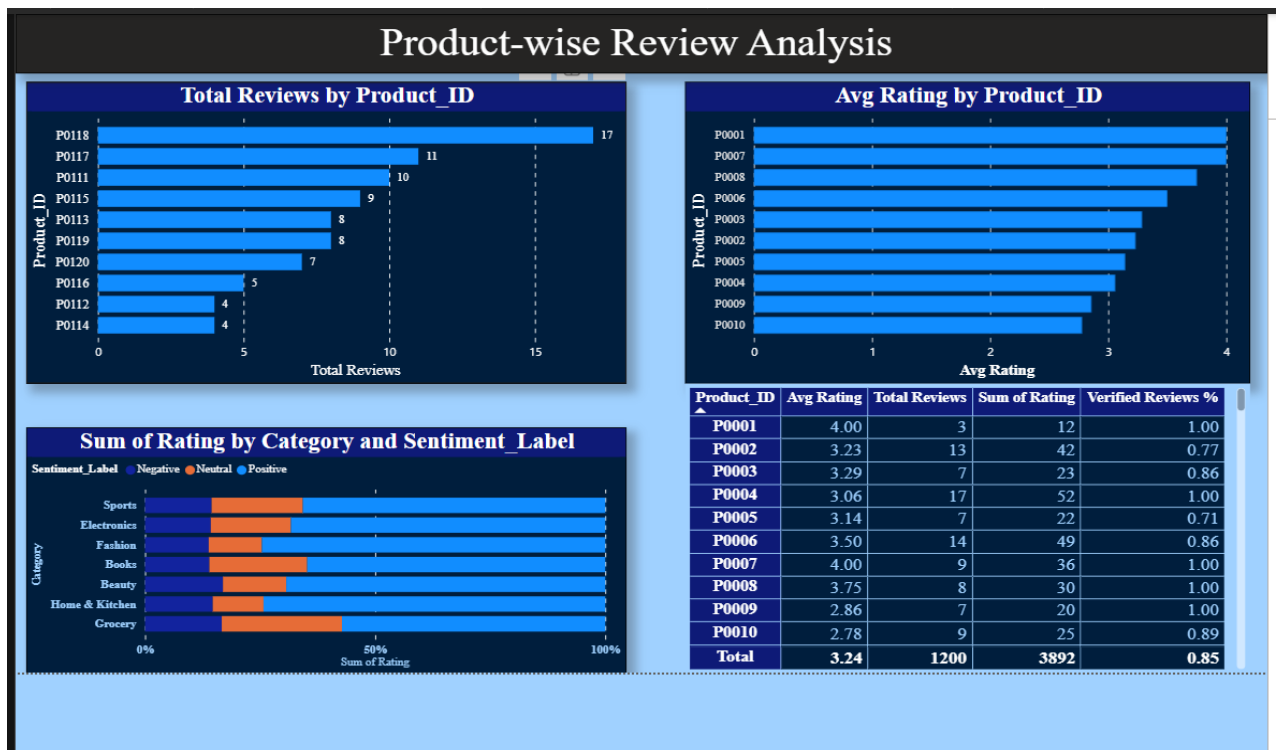


Figure 3.3: Product-wise Review Analysis

This dashboard focuses on analysing product-level performance based on reviews and ratings. It helps identify which products receive the most feedback and how customers rate individual products. The first visualization shows Total Reviews by Product ID. This bar chart displays the number of reviews received by each product. Products with higher review counts are generally more popular or more frequently purchased. This information helps businesses identify products that generate higher customer engagement. The second chart shows Average Rating by Product ID. This visualization compares the average ratings given to different products. Products with higher ratings indicate better customer satisfaction, while lower ratings may indicate potential issues with product quality, usability, or customer expectations. Another visualization in this dashboard is Sum of Rating by Category and Sentiment Label. This chart analyses customer sentiment across different product categories such as sports, electronics, fashion, books, beauty, home & kitchen, and grocery. The sentiment categories include positive, neutral, and negative reviews. This analysis helps businesses understand how different product categories perform in terms of customer satisfaction. Additionally, the dashboard includes a detailed data table that summarizes product information Verified Review Percentage This table provides a comprehensive.

3.4 Fourth page of the Dashboard

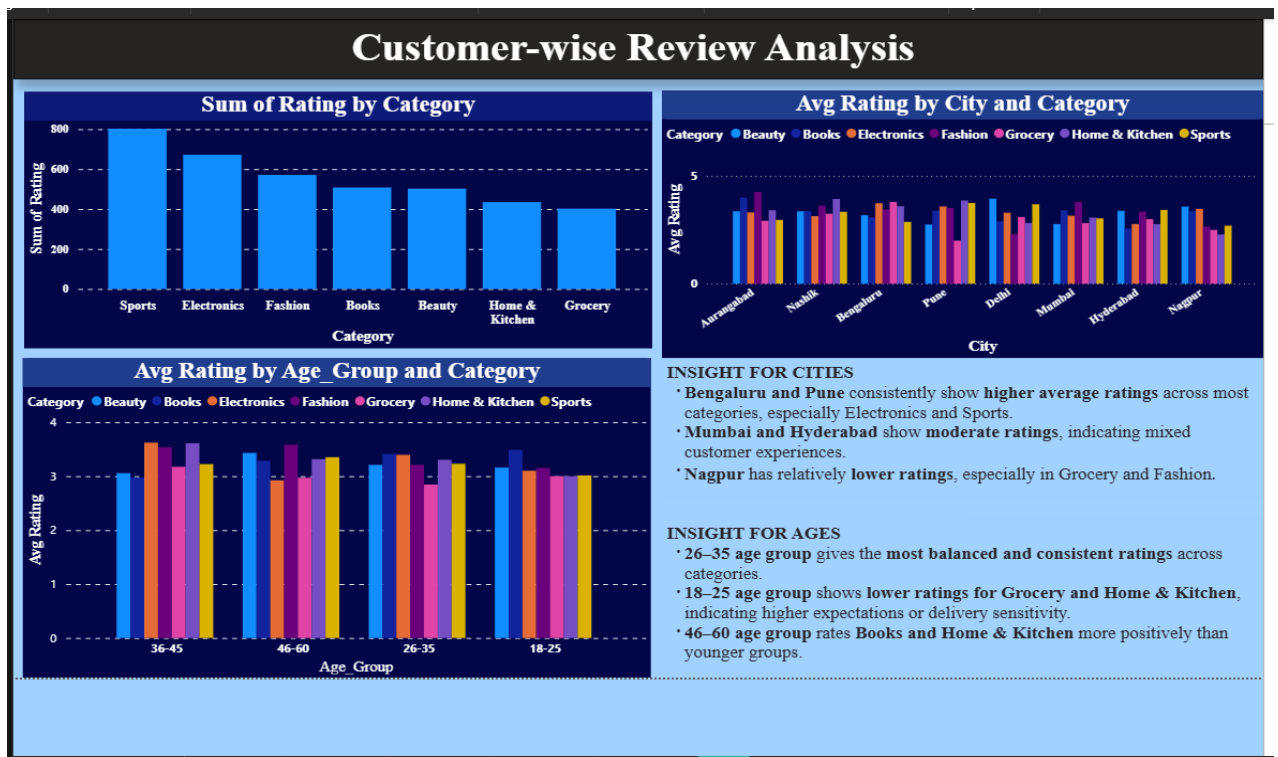


Figure 3.4: Customer-wise Review Analysis

This dashboard focuses on analysing customer behaviour and review patterns across different categories, cities, and age groups. It helps understand how different groups of customers perceive product quality and services. The first visualization shows Sum of Ratings by Category. This chart compares the total ratings received by each product category. Categories such as sports and electronics appear to have higher ratings, indicating stronger customer engagement and satisfaction. The second visualization presents Average Rating by City and Category. This chart analyses how customers from different cities rate products across various categories. From the analysis, it can be observed that cities such as Bengaluru and Pune tend to have higher average ratings, indicating higher customer satisfaction levels. In contrast, some cities show moderate or slightly lower ratings, suggesting differences in customer expectations or service experiences. Another important analysis presented in this dashboard is Average Rating by Age Group and Category. This visualization helps understand how different age groups evaluate products. The age groups analysed include: 18–25 years, 26–35 years, 36–45 years, 46–60 years. The analysis indicates that the 26–35 age group provides the most balanced ratings across categories, while younger age groups may have slightly higher expectations regarding service quality. Older

3.5 Fifth page of project Dashboard

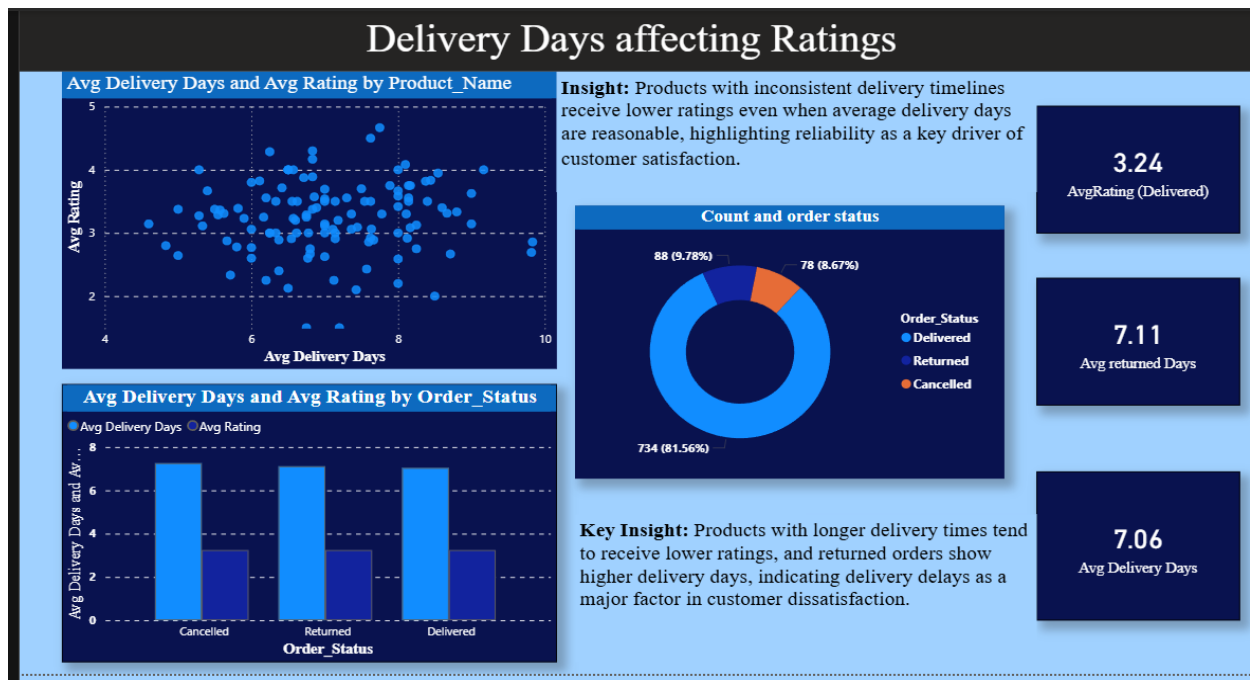


Figure 3.5: Delivery Days Affecting Customer Ratings

This dashboard analyses the relationship between **product delivery time and customer ratings**. Delivery performance is a critical factor influencing customer satisfaction in online shopping platforms. The first visualization is a **scatter plot showing Average Delivery Days versus Average Ratings by Product Name**. This chart helps identify whether longer delivery times negatively impact customer ratings. The distribution of points indicates that products with inconsistent or longer delivery timelines tend to receive slightly lower ratings from customers. Another visualization in the dashboard shows **Average Delivery Days and Average Rating by Order Status**, including categories such as: Delivered, Returned, Cancelled. This comparison helps analyse how delivery performance affects order outcomes and customer feedback. The dashboard also includes a **donut chart representing order status distribution**. The chart shows that the majority of orders are successfully delivered, while a smaller percentage of orders are returned or cancelled. Understanding this distribution helps businesses evaluate their delivery efficiency and order fulfillment performance. Additional KPI indicators displayed on the dashboard include:

- **Average Rating for Delivered Products:** 3.24
- **Average Returned Days:** 7.11
- **Average Delivery Days:** 7.06 **longer or inconsistent delivery times can lead to lower customer ratings and higher return rates.**

Chapter 4

**CONCLUSION AND
FUTURE SCOPE**

CONCLUSION

This project on *Data Analytics Using Python* shows how raw data can be converted into useful and meaningful information through proper analysis. In today's world, a large amount of data is generated every day, but it becomes valuable only when it is carefully processed and understood. This project highlights how data analytics helps in making better decisions by identifying patterns and trends in the data. Python was used as the main tool for this project because it is easy to use and has powerful libraries. Libraries like Pandas and NumPy helped in handling and analyzing data, while Matplotlib and Seaborn were used to create simple and clear visualizations. These tools made the entire process efficient and beginner-friendly.

The project followed important steps such as data collection, data cleaning, data analysis, and visualization. Data cleaning improved the quality of the dataset by removing errors and handling missing values. Exploratory Data Analysis helped in understanding the data better, while visualizations made the results easy to understand for everyone.

Through this project, useful insights were extracted from the dataset, showing how data can support decision-making. It also helped in improving knowledge of Python and developing analytical and problem-solving skills.

Overall, this project provided practical experience in working with real-world data and showed the importance of data analytics. It also builds a strong base for future learning in areas like machine learning and data science. Python proved to be a powerful and effective tool for data analysis, making it highly useful for both academic and professional growth.

FUTURE SCOPE

The future of *Data Analytics using Python* is very wide and continues to grow as more and more data is generated in different industries. Today, companies depend on data to make better decisions, and this demand will keep increasing. Because Python is easy to use and has powerful tools, it will remain an important language in the field of data analytics. In the future, this project can be improved by adding advanced techniques like predictive analytics and machine learning. Instead of only studying past data, it can be used to predict future trends, understand customer behavior, and identify risks. Python libraries can help in building models that make these predictions more accurate and useful. Another improvement is handling large amounts of data and real-time data. As data grows rapidly, basic methods may not be enough. This project can be expanded using big data technologies and cloud platforms so that large datasets can be processed quickly and efficiently, allowing faster decision-making. Automation is also an important future scope. Tasks like data cleaning, analysis, and reporting can be automated using Python. This will save time, reduce human errors, and make the process more efficient. Automatic systems can also generate regular reports without manual effort.

The project can also be enhanced by improving visualization. More advanced and interactive dashboards can be created to present data in a simple and clear way. This will help users easily understand and explore the data.

In addition, this project can be applied in many fields such as healthcare, finance, e-commerce, education, and manufacturing. Using real-world datasets from these areas will make the project more practical and useful.

Lastly, future improvements can focus on data security and ethics. Proper handling of sensitive data, privacy protection, and responsible use of data will be very important. Overall, the future scope of data analytics using Python is very promising. With new technologies and increasing data usage, this project can be developed into a more advanced and powerful system, creating good opportunities for learning and career growth.

Chapter 5

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